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MEMBRANE POTENTIALS IN THE ELECTROPLATES OF THE ELECTRIC EEL

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Ever since Walsh (1773) and Williamson (1775) suggested that the paralysing effect of touching certain fishes was due to an electric discharge, the properties of the electric organ have interested physicists and physiologists alike. It has long been recognized that in most species the electric organ is derived from muscle (Biedermann, 1898), but the exact way in which it has developed has remained in doubt. Several alternative theories have been put forward to explain the mechanism of the discharge (Fessard, 1946), but it has been difficult to decide between them, because the discharge has hitherto been recorded only with externally applied electrodes. Even when fragments of electric organ are used, such experiments cannot reveal the precise site of origin—at the cellular level—of the potential changes. We have therefore applied the type of internal microelectrode first developed by Graham & Gerard (1946), with modifications described by Nastuk & Hodgkin (1950), to investigate the membrane potentials in the individual electroplates of the electric eel, *Electrophorus electricus* L.

Our results, which have already been reported briefly (Keynes & Martins-Ferreira, 1952), show that across both faces of the electroplate there is normally a resting potential of the same size and polarity as that observed in other excitable tissues. During activity a large reversed action potential develops across the innervated face, while the potential across the non-nervous face remains virtually unaltered. Preliminary studies of the effects of factors such as the external sodium concentration on the sizes of these potentials suggest that they are produced by mechanisms similar to those responsible for the propagated impulse in nerve and muscle fibres. Considered in conjunction with the recent work of Fatt & Katz (1951*a*) on the motor end-plate, our experiments throw some light on the nature of the adaptation which has occurred in the electric organ.

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METHOD

As Fig. 3 shows, the electric organ is built up of a large number of electroplates, arranged in layers so that the potentials produced during activity are additive. It was difficult to cut slices of electric organ consisting of less than about three layers of electroplates, so that during the discharge the spaces between the layers did not remain at the potential of the surrounding fluid. The potential recorded between the tip of a single microelectrode and an indifferent non-polarizable electrode was hence not a reliable measure of the membrane potential. We therefore used two microelectrodes which could be manoeuvred independently, and determined the difference in potential between their tips, with the help of a direct-coupled differential pre-amplifier.

Material

Specimens of *Electrophorus*, varying in length between 0.4 and 1 m, were obtained from the Amazon region, and were kept at the laboratory in a shallow tank of fresh water. As long as the temperature was not allowed to fall below about 25° C, they survived very well. Each set of experiments necessitated the sacrifice of one fish, which was decapitated and then cut up into segments about 6 cm long. The segments were skinned, and the swimming muscles and Hunter's organ (see Fig. 3) were removed, leaving lumps of almost pure electric tissue, which were stored in Ringer's solution at room temperature. Immediately before making the measurements, the segment was sliced transversely, using a razor blade. The slices were roughly heart-shaped, and about 5 mm thick.

A few of the eels had been given injections of heparin about an hour before decapitation, in order to provide material for perfusion experiments (Chagas, Couceiro & Martins-Ferreira, 1951) which were being done simultaneously. This treatment had no detectable effect on the behaviour of their electroplates.

Apparatus

A diagram of the experimental lay-out is shown in Fig. 1. The microelectrodes were attached to probes containing R.C.A. Type 954 acorn pentodes connected as cathode-followers. One probe was mounted on a Zeiss micromanipulator, and the other on the vertical movement of a modified microscope stand. The cathode-follower circuit was identical, except for minor differences in component values, with that used by Nastuk & Hodgkin (1950). The differential pre-amplifier consisted of a pair of cathode-coupled triodes (6J7's with screen and suppressor grids strapped to the anodes), with a pentode (6J7) as the common cathode load. The output from the anode of one of the triodes was taken via a cathode-follower to the d.c. amplifier of a cathode-ray oscilloscope.

The microscope stand was provided with an attachable mechanical stage, fixed at right angles to the normal position. A Perspex dish was screwed rigidly to the stage. The slices of electric organ were placed in the dish, where they were held between a perforated silver plate and a grid of silver wire stretched across a Perspex frame, which could be lowered on to the slice by means of a small rack-and-pinion attached to the edge of the dish. Both plate and grid were coated with silver chloride. The dish could be flooded with Ringer's solution, and drained by a suction pump. The tissue slice was illuminated from below, using a low-power condenser lens, and insertion of the microelectrodes was observed with a Greenough binocular dissecting microscope on a long-arm stand.

Square-wave shocks, generated by a Grass electronic stimulator, were applied between the plate and the grid as shown in Fig. 2. A 4:1 step-down transformer could be switched into the circuit when it was necessary to reverse the polarity of the stimulus. A resistance of 1100 Ω was connected in series with the plate, in order to make the current effectively constant during the shock, whatever the resistance of the slice (which was usually between 3 and 15 Ω). It also prevented the slice

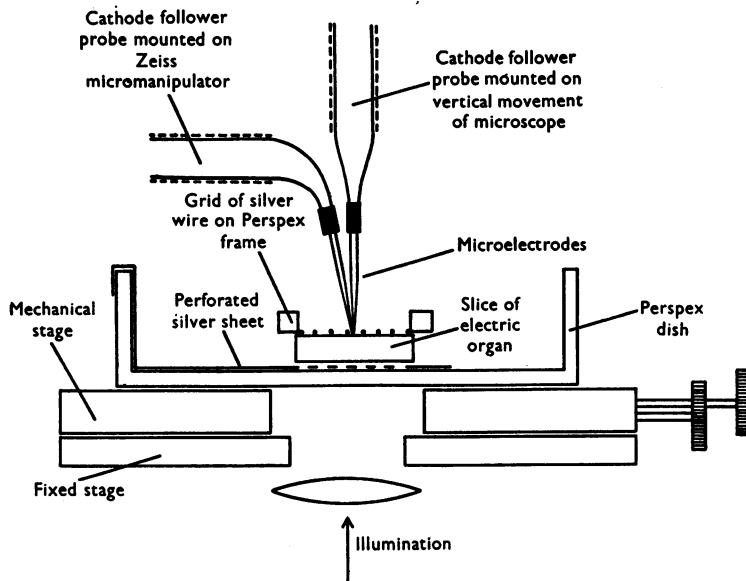


Fig. 1. Diagram of experimental arrangement for recording membrane potentials in slices of electric organ. Not to scale.

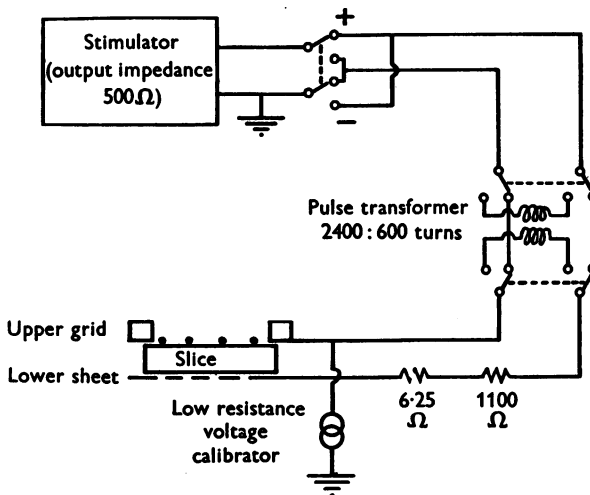


Fig. 2. Method of stimulating the electroplates. The output from the stimulator could be applied directly to the slice when positive pulses were needed, but had to be fed through a transformer (which slightly rounded the leading edge of the shock) when the polarity was reversed.

from being short-circuited by the secondary of the transformer. The applied current was measured by observing the potential developed across a monitor resistance of $6.25\ \Omega$, whose value (accurate to within $0.15\ \Omega$) was subsequently checked against a substandard. A low-resistance voltage calibrator of the type described by Hodgkin & Huxley (1945) was connected between the silver grid and earth. Time calibration was provided by a $1\ \text{kc/s}$ square-wave generator in the oscilloscope.

Solutions

A Ringer's solution based on the analyses of *Electrophorus* serum made by Hargreaves & Moreira (1949) was used throughout the work. Its composition is shown in Table 1. The table also shows the composition of a sodium-free dextrose solution calculated to be isotonic with the Ringer's solution. A similar solution containing $169\ \text{m.mole/l.}$ of choline chloride instead of the dextrose was used in some experiments.

TABLE 1. Composition of solutions (m.mole/l.)

<i>Electrophorus</i> Ringer			
NaCl	169	MgCl ₂	1.5
KCl	5	Na ₂ HPO ₄	1.2
CaCl ₂	3	NaH ₂ PO ₄	0.3
Isotonic dextrose solution			
Dextrose	300	MgCl ₂	1.5
KCl	2.3	K ₂ HPO ₄	1.2
CaCl ₂	3	KH ₂ PO ₄	0.3

Microelectrodes

Electrodes whose tip diameter was of the order of $0.5\ \mu$ were drawn out and filled with 3M-KCl by boiling at a reduced pressure. They were attached to the probes by short lengths of rubber tubing. Their characteristics were measured by the methods described by Nastuk & Hodgkin (1950). Their series resistance was usually between 4 and $15\ \text{M}\Omega$, and the total capacity of the input stage and microelectrode was of the order of $5\ \mu\text{F}$.

The main technical problem, which was never fully overcome, lay in finding microelectrodes which would give a good enough differential balance for transients. For steady potentials, the performance of the system was very satisfactory, its gain being over 5000 times greater for signals applied differentially than for the same signal applied to the tips of both microelectrodes. For transients, however, differences in the response times of the two channels, arising almost entirely from inequality of the microelectrode time constants, resulted in the appearance of out-of-balance artifacts. It was impracticable to improve the balance by adding small capacitances at one cathode-follower grid or the other, since the time constants varied with the depths to which the tips of the electrodes were lowered into the tissue slices. The electrodes were therefore sorted into matched pairs, but very exact agreement between them could not be demanded without rejecting too many. If with a square pulse whose rise time was under $1\ \mu\text{sec}$ applied to both electrodes, the artifact was less than 5 % of the output for the same signal fed in asymmetrically, the electrodes were acceptable; this test ensured that their time constants did not differ by more than 12 %. Since in practice the rates of rise and fall of the membrane potentials were smaller than those of the test pulses, this degree of balance was, for most purposes, sufficiently good to avoid serious errors. The artifact occurring during the rising phase of the spike was rarely greater than $2\ \text{mV}$, and certainly introduced no significant error into the determination of the height of its relatively flat peak. Only in investigating the voltage drop in different parts of the slice (see p. 326), where the differential balance of the system was more severely tested, would a better performance in this respect have been really advantageous.

Experimental procedure

The slice of electric organ was mounted in the dish, and rinsed with Ringer's solution. The dish was drained, and a suitable electroplate, i.e. one whose whole surface was exposed without any visible damage, was moved under the electrodes by manipulating the mechanical stage. The

standard method of recording the potentials when the nervous faces of the electroplates were uppermost, was as follows: One microelectrode was lowered into the jelly-like extracellular material until its tip was just above the surface of the electroplate. The second microelectrode was lowered alongside it until the tips were at the same level, and about 150μ apart. A base-line was then recorded. The second electrode was next lowered quickly a further 100μ , so as to penetrate into the electroplate. The size of the shock, a square pulse of duration 0.1 or 0.2 msec, was adjusted so that the rising phase of the spike was not obscured by the stimulus artifact, and another record was taken on the same frame as before. The electrode was finally withdrawn to its original position, and a third record was made. A number of records were taken with the two electrodes in different relative positions, or with the other side of the slice facing upwards, but the procedure was fundamentally similar.

As in other applications of this type of microelectrode (Nastuk & Hodgkin, 1950; Draper & Weidmann, 1951; Fatt & Katz, 1951*a*), a successful penetration was characterized by an abrupt change in the recorded potential as the tip of the electrode entered the cell, and any cases in which the potential altered sluggishly were rejected. Sometimes spurious variations in potential, up to about 20 mV in either direction, were observed while the electrodes were being lowered. Their origin was not explored in detail, but they probably arose from the combined effects of grid current and temporary obstruction of the electrode tip (causing a higher series resistance, and hence a positive-going variation in voltage), and from penetration into small nerve fibres in the layer of nerves, blood vessels, and connective tissue covering the nervous face of the electroplate (see Pl. 1). This layer was sufficiently tough to cause the failure of about one penetration in three. Some electroplates were penetrated repeatedly without trouble, but others were only penetrated at the second or third attempt, and a few entirely defied penetration. The connective tissue layer was almost always considerably dimpled before the electrode passed through it, and it was perhaps surprising that the tips were not often broken in the process. As might be expected, $5\text{ M}\Omega$ electrodes survived rather better than those with higher resistances and therefore finer tips, but as long as the shanks were straight and the electrodes were lowered along their own axis, a pair could generally be used for the whole of an experimental session without any serious decrease in resistance or visible breakage at the tip.

In experiments on temperature effects, a record was first taken at room temperature, and a rapid flow of cooled Ringer was then directed on to the surface of the slice. After 5 min, the temperatures of the stream of Ringer, of the surface of the slice, and of the Ringer accumulating in the dish appeared to be the same. The dish was therefore drained, and another record made. After a further 5 min wash in Ringer at room temperature, a final record was made. Experiments with sodium-free and potassium-rich solutions were done in a similar way, using longer periods of washing. Sometimes the flow was interrupted for just long enough to drain the dish and make a penetration, in order to observe intermediate stages in the development of the effects. While the solutions were running, the electrodes were raised in order to avoid damage to their tips. They were lowered into a different part of the electroplate for each penetration. It was inadvisable to leave them permanently lowered during the experiment, owing to the difficulty of allowing for amplifier drifts, and the danger of errors from the diffusion of KCl out of the tips.

Sources of error in the potential measurements

(a) The time constants of the microelectrodes and cathode-follower circuits were never greater than $75\mu\text{sec}$. Correction of typical records by the method of subtangent analysis (Lucas, 1912) showed that a time lag of this size would have reduced the amplitude of the recorded spike by much less than 1 mV, and would not have had any appreciable effect on its apparent duration.

(b) All records were measured against grids recorded by earthing one cathode-follower input, and increasing the injected calibration voltage in steps of 20 mV up to $\pm 200\text{ mV}$. The effects of curvature of the screen and non-linearity of the amplifier were thus eliminated.

(c) The negative grid current drawn by the cathode-follower tubes was $2 \times 10^{-11}\text{ A}$. This was well below the level at which it might have introduced measurable errors.

(d) Irreducible mains hum in the oscilloscope amplifier gave some trouble, causing appreciable vertical scatter in successive traces, and hence discrepancies between the two records of the baseline. In the records taken with the last eighteen eels, the average discrepancy was 1.6 mV. This source of error therefore contributed a standard deviation of about ± 1.4 mV to each value obtained for the resting potential. In the earlier experiments (using a different oscilloscope) the mains hum was still greater, its contribution being a standard deviation estimated as ± 4.0 mV.

(e) The strength of the shock had some influence on the amplitude of the action potential (see p. 332). It is unlikely that the spike size varied by more than ± 5 mV over the normal range of variation in shock size.

(f) As the microelectrodes were filled with 3M-KCl, the junction potentials between their tips and the intra- and extracellular material should have been rather small.

(g) Although our microelectrodes were probably somewhat larger at the tip than those used by Nastuk & Hodgkin (1950) and Draper & Weidmann (1951), they were not large enough to cause serious short-circuiting of the membrane potential at their point of entry. It was verified that an electrode of series resistance 0.7 M Ω (whose external tip diameter appeared to be of the order of 2.5μ) gave values for resting and action potentials which did not differ significantly from those obtained for the same electroplate with a 4 M Ω electrode, which was the lowest value normally used. Nor did 4 and 10 M Ω electrodes give different results. An even blunter electrode (5μ at the tip) gave the same action potential on being pushed right through the electroplate, but would not give a satisfactory indication of the resting potential. Attempts were made to see whether the resting potential recorded with one microelectrode was reduced when another electrode was inserted close by, but these failed because of movements of the electroplate while the second electrode was being pushed through the layer of nerve fibres and connective tissue.

(h) The recorded spike must have been slightly reduced by the voltage drop across the intra- and extracellular material due to current flowing during the discharge (see p. 326). In penetrating the nervous face the microelectrode was generally lowered about 100 μ , which would cause a loss of 2.7 mV (using the mean values given in Table 2). This estimate does not take into account the loss across the nervous face itself, which cannot easily be calculated. A spike current of 4.6 mA/cm² (see Table 2) flowing through the resting membrane resistance of 7.4 Ω cm² (see p. 333) would result in a loss of 34 mV, but a forty-fold drop in membrane resistance during activity, as was observed for squid axons by Cole & Curtis (1939), would reduce the effect to under 1 mV. Direct evidence that the voltage drop effect did not introduce large errors was obtained on one occasion by seeing whether the spike was affected by short-circuiting the protective resistance in series with the transformer secondary (see Fig. 2). As the output impedance of the stimulator was 500 Ω , this loaded the slice with a resistance of 500/4², or 31 Ω . Assuming the slice to have been 3 cm² in area, and four electroplates thick, so that its total discharge was 600 mV, the extra current drawn must have been of the order of 6 mA/cm². Nevertheless, the change in spike size was certainly under 1 mV.

(i) The electroplates chosen for penetration were always, as far as could be seen, intact and undamaged. In freshly prepared slices it was found that the electroplates which had been cut through at some point tended to give spikes which were smaller and shorter than normal; 1 mm from the cut end the resting potential was reduced to about half and the electroplate was inexcitable. When the condition of an electroplate deteriorated rapidly in the course of an experiment, the results were rejected, but they were often observed only over a short period, and it is probable that some slightly damaged electroplates passed undetected.

(j) From the scatter of the results in experiments where at least two separate penetrations were made into the same electroplate, the average standard deviation for a single penetration could be calculated. In twenty cases (for the last eighteen and not the first ten eels), it was ± 3.6 mV for the resting potential, and ± 4.3 mV for the action potential. There may therefore have been significant sources of variation other than those already listed, and perhaps genuine small differences in the membrane potentials at different points along the surface.

THE ANATOMY OF THE ELECTRIC ORGAN

It is necessary to describe very briefly the relevant anatomical features of the electric organ. For a fuller account, Biedermann (1898) and Couceiro & Akerman (1948) should be consulted. The electric tissue occupies most of the posterior four-fifths of the fish, as is indicated in Fig. 3. It is divided into three parts—the main organ, Hunter's organ, and the organ of Sachs. Each organ

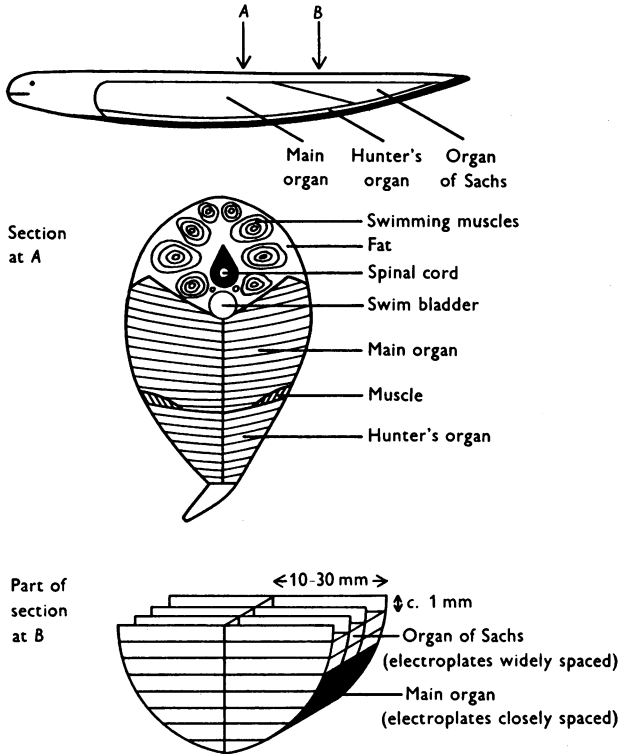


Fig. 3. Highly diagrammatic representation of the anatomy of the electric organ in *Electrophorus electricus*. The lower section, drawn in perspective, shows how the electroplates (whose faces lie in the plane of the paper) are arranged in columns.

is built up of a large number of electroplates, stacked in columns running parallel to the spinal cord, so that during activity a potential difference is developed between the head and tail of the eel. The individual electroplates are, very roughly, 100μ thick (in the longitudinal or antero-posterior direction), 1 mm wide (in the transverse or tangential direction), and 10-30 mm long (in the radial direction). In the main organ they are packed very closely together, but in the organ of Sachs they are spaced some distance apart, and the uppermost electroplates near the tail may have extracellular gaps over 2 mm wide (in the longitudinal direction) separating them. The photomicro-

graphs in Pl. 1, for which we are indebted to Dr A. Couceiro, show the appearance of the electroplates in sections cut vertically, parallel to the spinal cord. It can be seen that the afferent nerves and blood vessels are distributed only to the posterior faces of the electroplates, where the nerves terminate in endings on numerous small papillae which cover the whole surface. The anterior faces are studded with large irregularly shaped papillae, which project into the jelly-like extracellular material which fills the spaces between the electroplates. In the organ of Sachs the electroplates are somewhat wider, and the papillae are larger than in the main organ, but their structure is otherwise very similar.

The microelectrodes were lowered vertically on to the surface of slices cut perpendicular to the longitudinal axis of the fish. They thus penetrated the electroplates at right angles to the nervous faces. For good results it was essential to choose electroplates whose whole surface was exposed, but which had not been harmed in cutting the slice. This criterion could never be satisfied in slices cut from the main organ, owing to the close packing of the electroplates, but in a slice cut from the organ of Sachs there were generally a few electroplates suitably exposed. Almost all the experiments described here were therefore done with electroplates from the organ of Sachs. In fact, they owed much of their success to the convenient anatomy of this part of the electric organ.

RESULTS

The membrane potentials in an electroplate

Fig. 4 shows the potentials recorded when a microelectrode was lowered into an electroplate through the nervous (posterior) face, and then withdrawn. Record *A* was taken with both microelectrodes at the same level just above the electroplate, providing a base-line (with a superimposed stimulus artifact caused by the imperfect differential balance of the two channels for transients). Record *B* was then taken with one electrode lowered about 100μ so as to penetrate into the electroplate. As in a nerve or muscle fibre, the inside of the cell was about 85 mV negative with respect to the outside. Stimulation resulted in an action potential of about 150 mV, at the peak of which the membrane potential was reversed in the manner characteristic of many other excitable tissues (Hodgkin, 1951). On raising the electrode to its original position, record *C* was obtained, this being coincident with *A* except for a slight sideways displacement arising from random variation in time delay between the time-base synchronizing pulse and the stimulus.

Another series of records taken with the same electroplate is illustrated in Fig. 5. For records *A* and *B*, the two microelectrodes were in the same relative positions as before, but for *C* one of them was pushed right through the electroplate so as to emerge into the extracellular space below it. With the tips of the two electrodes thus on opposite sides of the electroplate, no steady

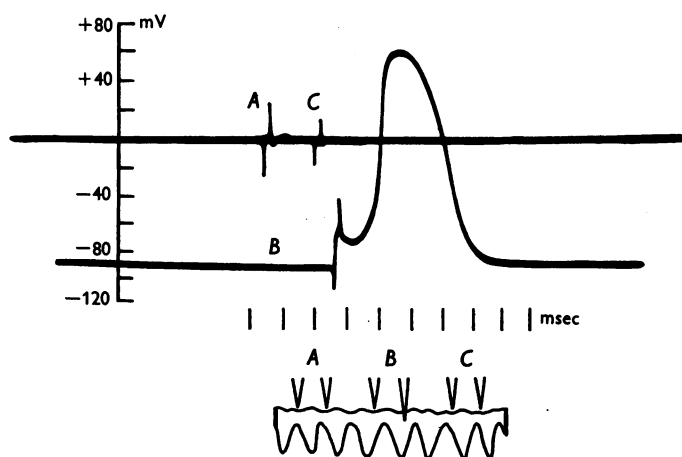


Fig. 4. Membrane potentials in an electroplate from the organ of Sachs, recorded by approaching through the nervous face. In the diagram of the stylized electroplate below, which shows the relative positions of the microelectrodes for the three records *A*, *B* and *C*, the non-nervous (anterior) face is drawn with large papillae. Temperature 27° C.

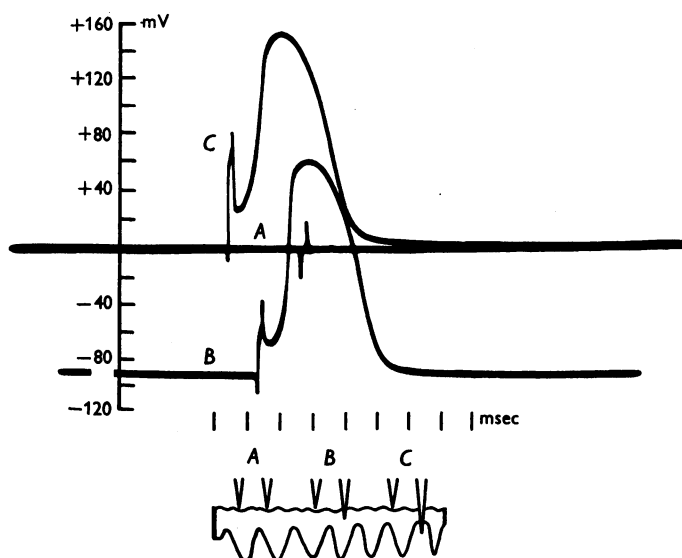


Fig. 5. Another set of records of the membrane potential in the same electroplate as Fig. 4. For the final record (*C*) the microelectrode was lowered one stage further than before, so as to transfix the electroplate and emerge into the extracellular space beneath the non-nervous face.

potential difference was recorded, but the action potential was almost the same size as it was across the nervous face alone (record *B*). It follows that the resting potentials across the two faces must have been equal, and that there was little or no action potential across the non-nervous face.

This conclusion is borne out by Fig. 6, which shows the result of inverting a slice of electric organ, and penetrating the electroplate from the other direction—through the non-nervous (anterior) face. Again the resting potential appeared when one electrode was inside and the other outside the

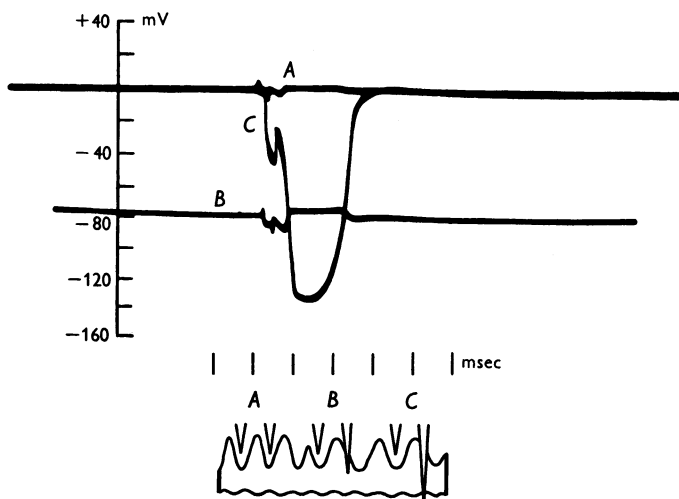


Fig. 6. Membrane potentials recorded by penetrating the non-nervous face of an electroplate. The action potential does not appear until the electroplate is transfixted (record *C*). Temperature 23° (not the same eel as Figs. 4 and 5).

cell, and disappeared when both were outside. During activity there was almost no potential change across the non-nervous face (record *B*), the action potential only being recorded, now upside-down in comparison with Fig. 4, when the tip of the movable electrode was beneath the electroplate (record *C*).

Examination of all the records in which electroplates had been transfixted by a microelectrode confirmed that there was generally no appreciable net potential difference across them. In one record the lower side of the electroplate was apparently 20 mV negative with respect to the upper side, but this was probably caused by a spurious shift of the base-line, and in the other records the figure rarely exceeded ± 6 mV. The average potential difference across the electroplates did not differ significantly from zero, since in twenty-one cases where the nervous face was uppermost, the lower side was negative by only 0.8 ± 1.3 mV (s.e. of mean), and in a further twenty-one records taken with inverted slices it was 1.2 ± 0.8 mV negative. It would not really have

been surprising if there had been a small potential across the electroplates, since whichever way the slices were orientated, their upper surfaces were bathed with Ringer's solution, in which the ionic concentrations may not have been exactly the same as in the extracellular spaces of the tissue. A diffusion potential might, therefore, have developed across the top electroplate. This effect was observed in a few experiments where the Ringer's solution was modified (see below), but evidently caused no easily measurable potentials under normal experimental conditions. It is justifiable to conclude that the resting potentials across the nervous and non-nervous faces of the electroplate were equal to within experimental error.

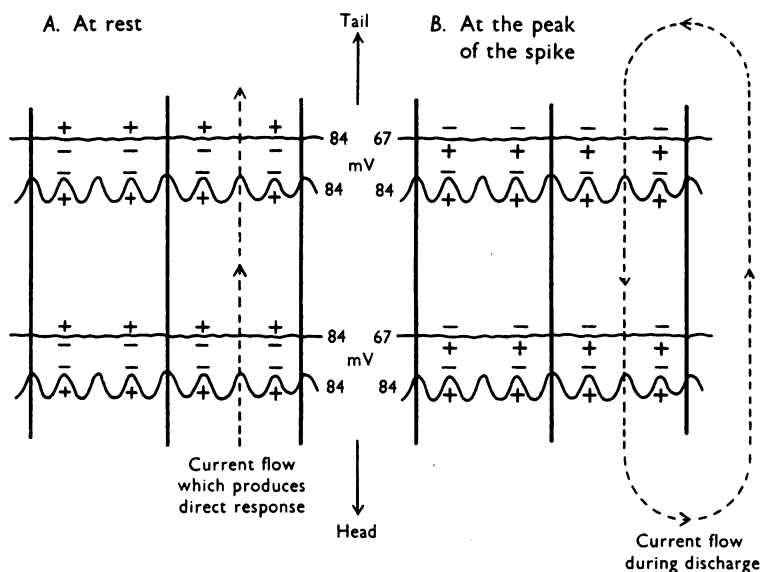


Fig. 7. Diagram illustrating the additive discharge of the electroplates. At rest (*A*) there is no net potential across the electroplates, but at the peak of the spike (*B*) all the potentials are in series, and the head of the eel becomes positive with respect to its tail. The figures are the overall averages obtained for electroplates in the organ of Sachs.

The voltage drop within the slice

It may be noticed in Fig. 5 that the action potential recorded across the whole electroplate (record *C*) was slightly smaller than it was across the nervous face alone (record *B*). This was often observed, the mean value of the difference in size of the action potential being 5.5 ± 1.2 mV (twenty-one records; for which a significance test gives $P < 0.001$). The discrepancy could be explained in two ways. There might be an action potential of this magnitude across the non-nervous face, as record *B* in Fig. 6 may suggest. Or, as the diagram in Fig. 7 shows, current flowing during the discharge might produce a voltage

drop across the intra- and extracellular material and across the passive resistance of the non-nervous face, which would offset the recorded action potential when the tip of one electrode was much above the other.

An attempt was made to discriminate between these two possibilities by investigating the contribution of the voltage drop effect. With increased amplifier gain, the vertical voltage drop in different parts of the slice could be determined by setting one electrode a known distance (generally 50 or 100 μ) above the other, and then measuring the potential difference developed between them during the spike. The wave-form of the pulse thus recorded was sometimes rather complex, owing to the poor transient balance of the recording system, but reasonably consistent results were obtained by measuring the potential only at the point corresponding in time to the flat-topped peak of the action potential. Whenever the non-nervous face lay between the two electrodes (as indicated by a change in resting potential), the recorded potential drop was slightly larger than for a corresponding depth of intra- or extracellular material. The difference gave a measure of the drop across the non-nervous face itself. A similar set of measurements was made with a short pulse of constant current flowing through the slice in the direction opposite to that which stimulated the electroplates. Knowing the size of the applied current and the area of the slice, and assuming that the current was distributed uniformly across the slice, the effective resistances of the cytoplasm of the electroplate, of the extracellular material, and of the non-nervous face, could next be calculated. Comparison of the voltage gradient for the spike and for the known constant current, across the same part of the slice, gave a figure for the current flowing during the spike. This was then multiplied by the resistance of the non-nervous face, in order to obtain the expected value of the voltage drop across it, if there were no active potential change.

The results of several experiments of this type are given in Table 2. There was apparently considerable variation in spike current, though some of this may have been due to experimental error, since the voltages were too small to be measured with great precision, and it was uncertain how far the assumption of uniform current distribution was justified. Nevertheless, the figures obtained for the specific resistances inside and outside the electroplates were fairly self-consistent, and fitted reasonably well with the specific resistance of the Ringer, which was calculated to be about 50 Ω cm at 25° C. Over an appreciable range of absolute sizes, the observed and calculated values of the voltage drop across the non-nervous face agreed rather well, neither of them being consistently smaller or greater than the other. This provides good evidence that there was no active potential change across the non-nervous face.

The results of two more incomplete experiments, omitted from Table 2, indicate that the intracellular specific resistance may not really be lower than the extracellular resistance, as Table 2 suggests. However, the cytoplasm of the electroplates does appear to be similar to the

axoplasm of frog myelinated nerve fibres in having an electrical resistance nearly equal to that of the body fluids, whereas other invertebrate nerves and frog muscle have relatively high internal resistances (Huxley & Stämpfli, 1951).

TABLE 2. The voltage drop across the non-nervous face of the electroplate during the action potential

Size of spike (mV)	Specific resistance		Spike current (mA/cm ²)	Resistance of non-nervous face (Ωcm ²)	Voltage drop in non-nervous face	
	Intracellular (Ωcm)	Extracellular (Ωcm)			Calculated (mV)	Observed (mV)
145	55	77	3.6	0.13	0.46	0.76
161	39	33	2.0	0.08	0.16	0.03
136	38	50	13.9	0.26	3.6	2.7
145	55	83	2.6	0.38	1.0	1.2
139	66	97	1.0	0.29	0.3	0.4
Mean	51	68	4.6	0.23	1.1	1.0

In most of the experiments, no record was kept of the actual amounts by which the electrodes were moved in order to penetrate the electroplates. However, the movements were certainly of the right order for the voltage drop effect to account both for the slight reduction in size of the spike when the electroplate was transfixated from the nervous side, and for the small positive-going change in membrane potential during the spike which is visible in Fig. 6, record *B*. Taking the average values from Table 2, a loss of 5.5 mV (see above) would correspond to lowering the electrode about 160 μ . In twenty-one penetrations through the non-nervous face (as Fig. 6), the mean apparent decrease in resting potential during the spike was 4.2 ± 1.0 mV, which could be caused by a movement of 110 μ .

The diagram in Fig. 7 summarizes the facts described so far, and shows how the additive discharge of a column of electroplates is achieved. In the resting state there is no net potential across the electroplates, however many are connected in series. During activity, the reversed potentials across the nervous faces are added to the unaltered potentials across the non-nervous faces, and each electroplate therefore contributes a voltage equal in size to its action potential.

Potentials in different parts of the electric organ

The organ of Sachs. Table 3 gives the mean values of the resting potential, of the action potential, and of its duration (measured at half the peak amplitude) in all the eels for which at least four different electroplates were penetrated through the nervous face. The corresponding figures for a further eleven eels, in each of which not more than three records were obtained, were: R.P. 82 mV, A.P. 146 mV, and duration 2.2 msec. Averaging the two sets of figures, and allowing for the fact that the spike was reduced about 3 mV below its true size by the voltage drop effect (see p. 320), the best overall estimate for the membrane potentials in the organ of Sachs at 24° C was: R.P. 84 mV,

A.P. 151 mV, and duration 2.3 msec. The averages for penetrations through the non-nervous face (also in eleven eels)—R.P. 83 mV, A.P. 141 mV, and duration 2.0 msec—agreed well, since the total correction for the voltage drop effect was about 7 mV when recording in this fashion. The table shows the standard error of each figure, and it is clear that there was some significant variation from one eel to another. However, the variation in membrane potentials was not very extensive, and it was only the durations which differed appreciably between individual fishes. Part of the variability in duration was probably caused by temperature differences; taking its Q_{10} as 2.5 (see below), and correcting all the results to 24° C, the range of durations in all twenty-two eels was reduced to 1.4–3.1 msec.

TABLE 3. Average sizes of resting and action potentials in electroplates of Sachs's organ

Eel no.	Temp. (° C)	No. of records	Resting potential (mV)	Action potential (mV)	Overshoot (mV)	Duration (msec)
7H	22	4	88 ± 1	145 ± 3	56 ± 3	2.1 ± 0.05
8	21	4	100 ± 2	158 ± 5	59 ± 5	—
10	24	4	82 ± 2	152 ± 8	70 ± 5	—
12	23	6	82 ± 2	151 ± 5	69 ± 4	2.2 ± 0.16
14	24	13	88 ± 1	155 ± 3	68 ± 2	1.7 ± 0.05
15H	24	17	88 ± 2	154 ± 3	66 ± 2	2.9 ± 0.09
19	22	7	88 ± 2	153 ± 3	64 ± 2	3.7 ± 0.16
20H	23	5	81 ± 1	152 ± 3	72 ± 2	2.9 ± 0.18
21	24	15	76 ± 2	137 ± 3	61 ± 2	2.2 ± 0.11
27	27	5	90 ± 4	144 ± 4	54 ± 4	1.3 ± 0.03
28	25	5	83 ± 4	152 ± 8	69 ± 4	2.5 ± 0.12
Mean	24	—	86 ± 1.9	150 ± 1.8	64 ± 1.8	2.4

All these measurements were made by penetrating the electroplate through the nervous face. For the first six eels, non-standard techniques were used, and the results were rejected altogether. Eels marked H had been injected with heparin before starting the experiment. The duration was measured at half the peak amplitude of the action potential. Standard errors of the means for each eel are given, and also the standard errors of the overall means (which are not weighted).

The greatest action potential recorded in any electroplate was 180 mV in the first electroplate of eel no. 14 to be penetrated; its resting potential was 90 mV. This may seem a surprisingly large value, but there was no reason to doubt its genuineness, and most of the possible errors in determining it would have made it too small rather than too large (see 'Method'). This record was made sooner after the death of the fish (about 20 min) than any other in the whole series, and it is conceivable that there was a sharp initial decline in the size of the spike, which thereafter remained steady for some time. Eel no. 14 gave a mean of 168 ± 5 mV (four records) 0.5 hr after decapitation, and 153 ± 1.7 mV (four records) some 3.3 hr later (which is a significant change, $P=0.03$). The fall in action potential with time was sometimes greater than this, and sometimes there was less change (eel no. 15H gave 159 ± 4 mV in four of its electroplates 0.5 hr after decapitation, and precisely the same result in another four penetrations made 3.7 hr later).

In any given slice through the organ of Sachs there was always appreciable variation in the longitudinal spacing of the electroplates. In a large eel, for instance, the distance between the electroplates might be 2.3 mm in the upper part of the organ (nearest the spinal cord), and only 0.8 mm in the lowest part, with intermediate spacings in between. It seemed possible that there would be some correlation between spacing and size of action potential, but no convincing evidence was obtained that the size of the spikes was related to the position of the electroplates. Nor was there any obvious connexion between the total area of the electroplates and their spike size. These questions were not investigated thoroughly enough for the existence of such correlations to be ruled out, but it was quite clear that any variations in action potential were much smaller than the variations in spacing and in area. If all the electroplates discharge simultaneously, there must be marked discrepancies between the voltages generated by different regions of the electric organ, and a rather complicated pattern of current flow within it.

The main organ. Once the technique of penetrating the electroplates in the organ of Sachs had been well established, it proved possible to make satisfactory records from the electroplates in the main organ. Electrode *A* was first lowered into the slice, at whichever end wholly undamaged electroplates were nearest the surface, until it was below at least one electroplate giving large resting and action potentials. Electrode *B* was then lowered alongside it, until there was no potential difference between them during stimulation, and they were therefore at the same level. After recording a base-line, electrode *B* was lowered one step further still, the potentials were recorded, and the electrode was withdrawn. With this procedure it was, of course, impossible to observe the electrode tips visually, but as the sequence of potential changes was exactly the same as in the organ of Sachs, there could be no doubt as to the relative positions of the two electrodes. Table 4 summarizes the results from the only two eels in which the membrane potentials in the main organ were examined systematically. As Fig. 8 shows, the action potentials had the same general appearance as those in Sachs's organ, but were significantly shorter, their mean duration being 1.5 msec as compared with 2.2 msec. This helps to explain the surprising shortness of the pulses produced by a whole eel (Albe-Fessard, Chagas, Couceiro & Fessard, 1951), which would correspond to the discharge of the main organ, not of Sachs's organ, although the individual electroplates must, all the same, be extraordinarily well synchronized. The resting and action potentials were, on the whole, slightly smaller than in the organ of Sachs, though the differences were not statistically significant in every case. The average size of the spikes in the three groups of records was 127 mV, which agrees well with the estimates of 100–160 mV obtained by Cox, Coates & Brown (1945) from measurements of the total voltage and total number of electroplates in intact eels.

Hunter's organ. The discharge of Hunter's organ, in which the electroplates are spaced closely as in the main organ, was only examined on one occasion. One electroplate gave a resting potential of 74 mV and an action potential of 141 mV, and another an action potential of 128 mV. The spikes were the same shape as before.

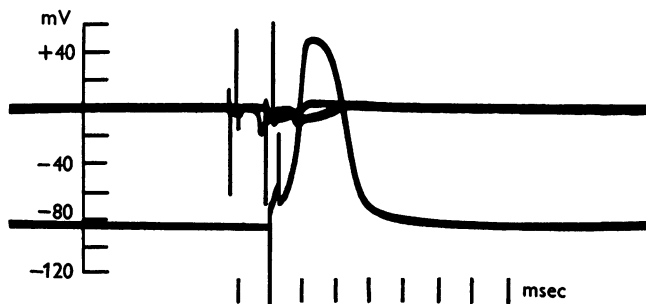


Fig. 8. An action potential recorded in the main electric organ. The electroplate was penetrated through its nervous face (as in Fig. 4). Electroplate spacing 0.6 mm. Temperature 24° C.

TABLE 4. Comparison of potentials in main and Sachs's organs

Region	No. of records	Resting potential (mV)	Action potential (mV)	Duration (msec)	Electroplate spacing (mm)
{End of main organ	4	80 ± 4	132 ± 11	1.5 ± 0.06	0.6 }
{Beginning Sachs's organ	3	87 ± 4	152 ± 5	2.1 ± 0.09	1.7 }
Centre of main organ	4	64 ± 4	131 ± 6	1.4 ± 0.12	0.1
{End of main organ	9	68 ± 3	118 ± 6	1.5 ± 0.06	0.2 }
{Beginning Sachs's organ	6	81 ± 2	145 ± 4	2.3 ± 0.14	0.7 }
Pure Sachs's organ	9	72 ± 3	131 ± 4	2.1 ± 0.17	1.0

Figures bracketed together were obtained with the same slice. The first pair were from one eel, the remainder from another. All the measurements were made at a room temperature of 24° C. The standard errors of the means are given.

The effect of temperature

Fig. 9 shows how the action potential changed when the electroplate was cooled, and Table 5 gives numerical results for the temperature effects in this and three other electroplates. These measurements were made at an early stage in the investigation when imperfections in technique were still causing an unduly large scatter in the resting potential records, the standard deviation of each figure being about ± 5 mV. The slight reduction in resting potential at the lower temperature was, therefore, not statistically significant. The action potentials were determined somewhat more reliably, and decreased appreciably on cooling. Their average Q_{10} , 1.08 ± 0.04 , was not significantly different from the value corresponding to a direct proportionality to absolute temperature. In contrast, the duration of the spike had a large negative temperature coefficient, and increased to more than 8 msec at 6° C. This lengthening was apparently not due simply to a slowing of the rising and falling phases since

the maximum slopes had rather smaller Q_{10} 's than the duration. Hodgkin & Katz (1949) reported similar temperature effects for squid axons, where the sizes of the potentials did not vary much, and the rate of restoration of the resting potential at the end of the spike had the largest Q_{10} .

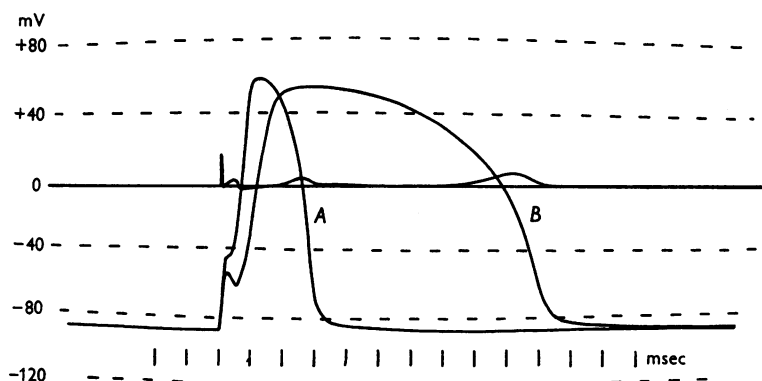


Fig. 9. Superimposed tracings showing the effect on the action potential of cooling the electroplate. Curve *A* is drawn as the mean of potentials recorded through the nervous face (as in Fig. 4) at 21.5° C, before and after cooling. The two values for the resting potential were 88 and 80 mV respectively, and the spikes were virtually identical. Curve *B* was recorded at 7.5° C.

TABLE 5. The effect of temperature on the membrane potentials

Temp. (° C)	Resting potential (mV)	Action potential (mV)	Duration (msec)	Maximum rate (V/sec) of		Q_{10}			
				Rise	Fall	A.P.	Duration	Rise	Fall
{ 23	80.5	143	1.7	251	143	1.03	2.33	1.50	1.59
{ 12	80	138	4.3	160	86				
{ 22	91.5	134.5	2.0	158	120	1.11	2.41	1.60	1.73
{ 6	83	115	8.2	75	50				
{ 21	91.5	140.5	2.3	179	108	1.17	2.52	—	2.04
{ 11	89	120	5.8	—	53				
{ 21.5	84	148.5	2.15	188	172	1.01	2.64	1.43	1.81
{ 7.5	84	146	8.4	114	75				
Mean 22	88	142	2.0	194	136	1.08 ± 0.04	2.48 ± 0.07	1.51 ± 0.05	1.79 ± 0.09

Each figure at the upper temperature is the mean of two measurements, made before and after cooling. Durations were measured at half the peak amplitude. Maximum rates of rise and fall were measured graphically from the action potential records. In one case the stimulus artifact obscured too much of the rising phase for the rate of rise to be measured. The standard errors of the mean Q_{10} 's are given.

Excitation of the nervous face

The electroplates in slices of electric organ from *Electrophorus* can be stimulated in two ways—either directly, with a latency often under 0.1 msec, or indirectly, through stimulation of the nerve endings, with a latency of at

least 1.5 msec (Albe-Fessard, Chagas & Martins-Ferreira, 1951). The direct response was used in all our experiments, this being elicited by brief pulses of current passed through the slice in the antero-posterior direction, but not by the flow of current in the opposite direction. We did not perform any critical experiments on the mechanism of excitation in the electroplates, although the type of technique evolved would be suitable for further investigation of this very interesting question. But we made a few relevant observations, largely qualitative in nature, which are worthy of mention here.

It will be obvious on referring to Fig. 7 that the current which caused direct excitation was that which depolarized the nervous face of the electroplate. This mode of stimulation was thus analogous to the usual method of stimulating a nerve or muscle fibre by applying a cathode externally. The analogy

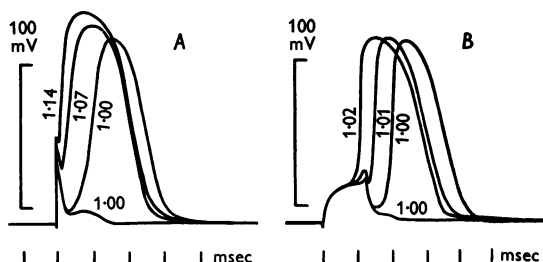


Fig. 10. Superimposed tracings of the responses of an electroplate when the size of stimulus was varied, recorded through the nervous face. The figures show relative shock strengths. *A*, 0.02 msec shock, threshold about 17 mA/cm². *B*, 1.2 msec shock, threshold about 1.5 mA/cm². Temperature 25° C. In each case both types of response at threshold were recorded without touching the shock strength control.

extended somewhat further, since the nervous face responded in a characteristic all-or-none fashion, as may be seen in Fig. 10, which shows the effect of varying the stimulus strength, for pulses 0.02 and 1.2 msec in duration. As the spike arose from synchronous (or almost synchronous) activity of the whole surface of the electroplate, the main features of these records should be compared with those of the non-propagated membrane action potentials obtained in squid axons by Hodgkin, Huxley & Katz (1952), rather than with normal propagated action potentials. The stimulus strength could be set at a level where the electroplate sometimes gave a subthreshold response resembling the local response observed in many other excitable tissues, and sometimes, after a delay of several milliseconds, a full-sized spike. By applying a larger shock, the delay could then be reduced virtually to zero without much change in the appearance of the spike. However, the all-or-none law was often not perfectly obeyed, since the spike showed a tendency, particularly when the electroplate seemed to be in poor condition, to be slightly smaller and shorter close to threshold. This effect cannot have arisen from distortion of the record by the

stimulus artifact, since by the time the peak of the spike was reached, the artifact was very small. It may possibly have been a consequence of the complex folding and corrugation of the electroplate surface, which must have resulted in a non-uniform distribution of the stimulating current, and hence incomplete excitation of the membrane at threshold. But if this idea were correct, it appears likely that the threshold response would be asynchronous, and therefore lengthened, instead of being shorter, as Fig. 10 shows it to be. For the present, the departures from ideal all-or-none behaviour must be left unexplained.

Using shocks over 10 msec in duration, the average voltage drop produced across the nervous face at threshold was measured, and found to be 11.4 mV (mean of seven observations ranging from 10 to 13 mV). The corresponding current was 1.65 mA/cm², and 1.0 mV was subtracted from the observed values to allow for the voltage drop across 100 μ of extra- and intracellular tissue (see p. 327). This figure is comparable with the critical depolarization of 12–15 mV necessary to excite the squid axon membrane (Hodgkin *et al.* 1952). When shorter shocks were used, the initial depolarization required at threshold was considerably greater, as may be observed in Fig. 10. When the shock was very long (20 or 50 msec), repetitive responses were sometimes obtained as soon as the depolarization exceeded the critical value.

On several occasions the indirect response was elicited, in order to see how it compared with the direct response. Fig. 11 shows a typical picture of the direct and indirect responses in the same electroplate. In confirmation of Albe-Fessard, Chagas & Martins-Ferreira (1951), it was noted that the indirect response required longer and larger shocks, that it arose with an irreducible latency of about 1.5 msec, that it was more easily fatigued (slices which had been maltreated, or left to soak in Ringer's solution for long periods, sometimes gave an almost unimpaired direct response when they would give no indirect response at all), and that it showed marked facilitation effects (it was most easily obtained by stimulating at a repetition rate of at least 30/sec). No initial steps of the type recorded by Fatt & Katz (1951*a*) near motor end-plates in frog muscle were ever seen, but no serious effort was made to search for end-plate potentials, and it is possible that they might have been found if nerve-organ preparations had been used, and if trouble had been taken to insert one of the microelectrodes close to a nerve ending.

Membrane characteristics

The effective resistance of the nervous face was determined in a number of experiments similar to (but often less complete than) that described on p. 326, i.e. by measuring the steady component of the voltage drop across it when pulses of current were applied to the slice. The average result was 7.4 Ω cm² (eleven values ranging from 4.1 to 12.8 Ω cm²). Most of the measurements were made with 1 msec pulses of current flowing in the postero-anterior direction, the current densities being between 1 and 7 mA/cm². On several occasions the voltage drop was also measured with just subthreshold currents flowing in the reverse direction. No evidence was found of

any rectification effects, the membrane resistance being the same to within experimental error for currents up to about 1.5 mA/cm^2 flowing in either direction. In cases where the specific resistance of the cytoplasm and extracellular material was not determined at the same time, a correction of $0.6 \Omega\text{cm}^2$ was subtracted from the result, in order to allow for a vertical distance of 100μ between the electrode tips (and a mean specific resistance of $60 \Omega\text{cm}$, from Table 2). There were two further sources of error for which reliable compensation could not easily be made, but which tended to cancel one another out. One was the presence of connective tissue septa between the columns of electroplates, which occupied about 10 % of the total area, and which probably

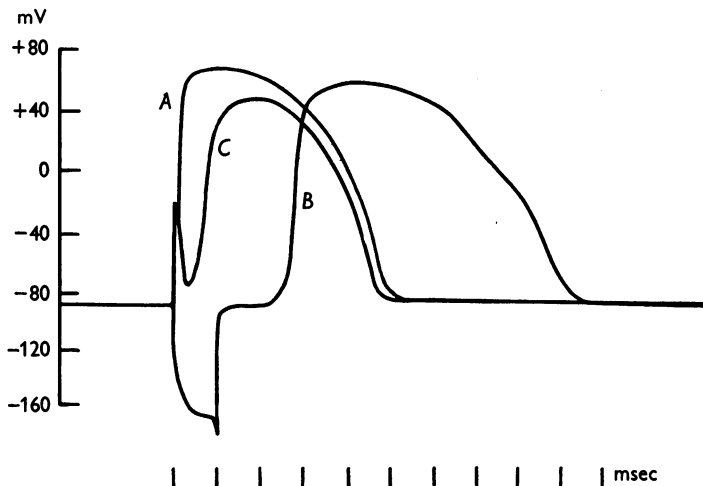


Fig. 11. Superimposed tracings of direct and indirect responses for an electroplate in the tail end of Sachs's organ, recorded through the nervous face. *A*, direct response, 0.1 msec shock, cathodal, about 1/sec; *B*, indirect response, 1 msec shock, anodal, 30/sec; *C*, direct response, 0.1 msec shock, cathodal, 30/sec. Strength of shock the same in each case. Temperature 23°C . *C* was recorded last, and the spike was probably somewhat reduced by fatigue and a rise in threshold. The hump on the descending phase of the indirect response (*B*) was not always seen, and in some other records the direct and indirect responses were exactly the same in size and in duration.

had a higher specific resistance than the rest of the slice. The other was the existence of short-circuits within the slice, which allowed a mean current of 4.6 mA/cm^2 to flow during the spike (see Table 2). It was calculated that the apparent membrane resistance was probably too low by 10–30 %, the error being greatest if the short-circuit path was assumed to lie across each individual electroplate, and least if it represented only a leakage round the outside of the slice. Taking all these errors into account, it seems likely that the true membrane resistance was between 8 and $10 \Omega\text{cm}^2$.

When a rectangular current pulse was applied to the slice, the voltage across the nervous face rose exponentially to a steady value, as can be seen in tracing *B*, Fig. 10. In seven experiments in which records were made with a fast sweep, the time constant of the exponential varied between 60 and $200 \mu\text{sec}$ (only the latter part of the build-up was measured, since the inclusion of some series resistance between the electrodes added a vertical portion to the beginning). Knowing the membrane resistance in each case, the average membrane capacity was calculated to be $15.6 \mu\text{F/cm}^2$ (with a range of $10\text{--}21 \mu\text{F/cm}^2$). The reliability of this figure is again affected by assumptions as to the distribution of the short-circuit resistance, but it should be correct to within 30 %. It is high compared with the values of the order of $1 \mu\text{F/cm}^2$ obtained for nerve membranes (Hodgkin, 1951), but is not out of line with the figures reported for the electric organ of *Torpedo* ($5 \mu\text{F/cm}^2$,

Albe-Fessard, 1950*b*; Albe-Fessard, Chagas & Fessard, 1949) and for muscle ($6\mu\text{F}/\text{cm}^2$ in frog muscle and $40\mu\text{F}/\text{cm}^2$ in crab muscle, Fatt & Katz, 1951*a, b*). It must also be remembered that the true area of the membrane is considerably greater than the area assumed in these calculations, since it is extensively folded.

Figures for the resistance of the non-nervous face have already been given in Table 2. The average value obtained was $0.23\ \Omega\text{cm}^2$. There was no sign of an exponential change in potential across this face, but the voltages involved were too small for any final conclusions to be reached with regard to its capacity. This remarkably low figure has obvious functional importance, and must partly be achieved by the increase in surface area resulting from the presence of exceptionally large papillae on the non-nervous face. Examination of Pl. 1 suggests that the papillae may expand the available surface by a factor of about five.

Evidence has been given earlier (see p. 320) that the resistance of the nervous face was appreciably reduced during the spike. This question was not examined in detail, as it would have required a double pulse generator. However, the disappearance of the back edge of the applied current pulse when the shock strength was 1.02 in tracing *B*, Fig. 10, provides a further demonstration that the membrane resistance must have been low by the time the peak of the pulse was reached. Cox *et al.* (1945) have obtained indirect evidence that the resistance of the main organ falls during the discharge.

The effect of applying sodium-free solutions

It has been shown (see Hodgkin, 1951) that many types of nerve and muscle fibre are rendered reversibly inexcitable when the sodium ions in the external medium are replaced by sugar or by an inert cation such as choline. It was therefore of interest to see whether the electroplates were similarly affected.

The experiment illustrated in Fig. 12, and those summarized in Table 6, showed that when the nervous face of an electroplate was washed with an isotonic dextrose solution, the action potential was abolished without any great change in the resting potential. But diffusion effects caused the inexcitability to develop rather slowly, and prevented the results from being as clear-cut as might have been wished. It can be seen in Fig. 12 that 6 min elapsed before the spike was reduced to quarter of its original height, and that after 12 min traces of an active response still persisted. Much of this delay must have resulted from the appreciable time necessary for sodium ions to diffuse out of the layer of jelly-like material in which the electroplates were embedded, which could not, of course, be stirred. Some idea of the maximum speed of diffusion can be gained from Fig. 13, which shows how rapidly the action potential would drop if the active membrane behaved as a sodium electrode, as it does in certain other excitable tissues (Hodgkin, 1951), and were protected by unstirred layers of various depths. In the experiment of Fig. 12 the unstirred layer was of the order of 400μ deep, so that the spike declined at very roughly the theoretical rate. With one exception, the other experiments also showed a very approximate agreement between depth and blocking time, the rate of fall of the spike sometimes being slightly less than Fig. 13 would have indicated, perhaps because the connective tissue overlying the surface of the electroplate offered further hindrance to the free diffusion of sodium ions. In the fifth experiment listed in Table 6, there was still a 90 mV

spike after 26 min of exposure to a dextrose solution, although the depth was not more than 500μ . It can only be supposed that the connective tissue must have been unusually impervious on this occasion. In every case, the recovery of the spike on restoration of normal Ringer was noticeably quicker than its previous decline—in Fig. 12 it was virtually complete in 3 min. However, this is exactly what the diffusion calculations predicted, as is shown by the right-hand set of curves in Fig. 13.

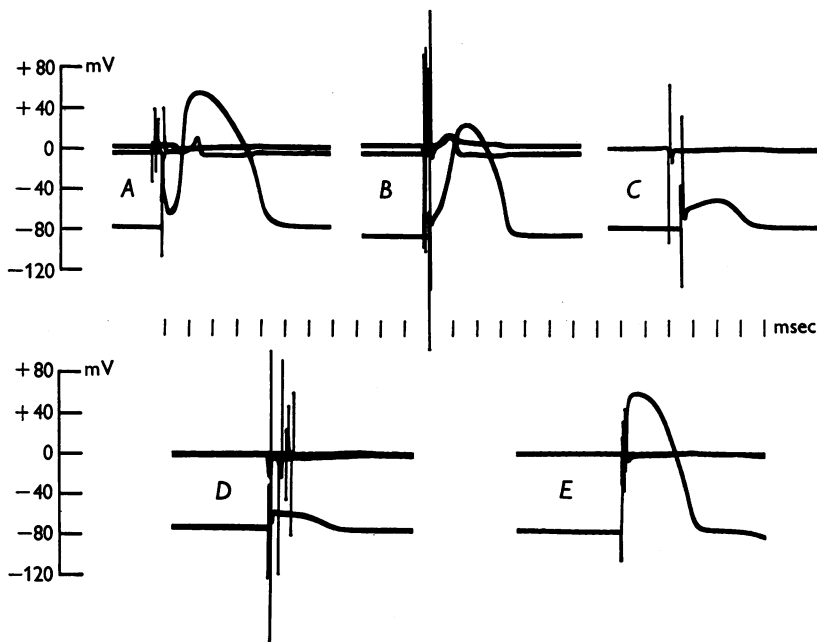


Fig. 12. The effect of washing the nervous face of an electroplate with an isotonic dextrose solution. *A*, initial response in normal Ringer; *B*, *C* and *D*, after 3, 6 and 12 min treatment with dextrose; *E*, recovery of response after 3 min in normal Ringer. Depth of unstirred layer about 400μ . Temperature 24°C . Shock in each case 0.1 msec , 7.3 mA/cm^2 . Responses in *C* and *D* could not be increased by applying larger shocks. In other experiments the response was abolished more completely than in *D*. Transient balance of microelectrodes was rather worse than usual, and transient artifacts were increased in dextrose because of its higher electrical resistance.

When an isotonic choline solution was applied to the nervous face instead of dextrose, the electroplates again became inexcitable, but now irreversibly so (Table 6). In addition, there was a significant reduction in the resting potential. As there is evidence that the nerve endings on the electroplate are cholinergic (see Discussion), and since choline irreversibly abolishes the excitability of the motor end-plates in frog muscle after first producing a depolarization which only slowly disappears (Fatt, 1950), it is plausible to suggest that the action of the choline solution was partly to depolarize the

electroplates in the region of the nerve endings, and so to block the whole of the nervous face, even after replacement of the external sodium. In support of this idea, we found that application to the nervous face of freshly prepared Ringer containing 11 m.mole/l. acetylcholine chloride also reduced the resting potential by half, and destroyed excitability.

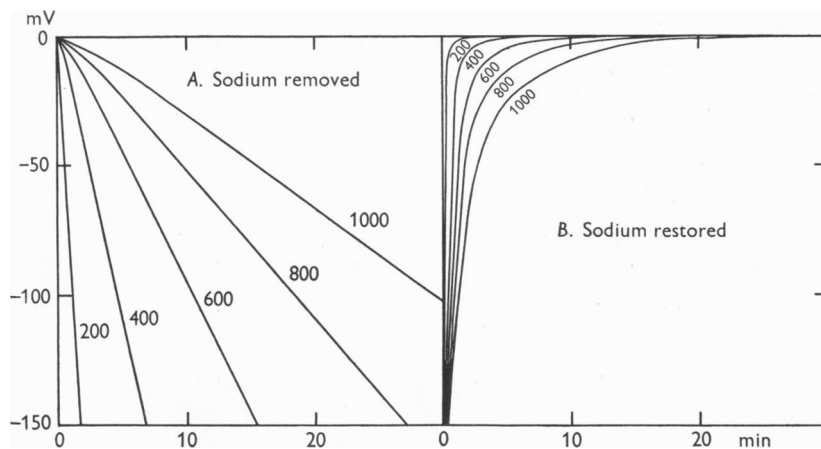


Fig. 13. Theoretical reduction in membrane potential on altering external sodium concentration, for a membrane which behaves as a sodium electrode and has an unstirred layer above it. The figures against the curves indicate the thickness of the unstirred layer in μ . Calculated from the diffusion equations given by Hill (1929) for a plane sheet, taking the diffusion constant for Na^+ ions as $1.0 \times 10^{-5} \text{ cm}^2/\text{sec}$, and assuming that

Drop in potential $= 58 \text{ mV} \log_{10} \frac{[\text{Na}]_0}{[\text{Na}]_t}$, where $[\text{Na}]_0$ is initial (and final) external sodium concentration, and $[\text{Na}]_t$ is the sodium concentration immediately outside the membrane at time t . Recovery curves calculated for zero initial sodium concentration.

In contrast to the results just described, application of either dextrose or choline solutions to the non-nervous face caused little alteration of the action potential. As may be seen from Table 6, it fell slowly after about half an hour, but this might have been a consequence of damage caused by repeated penetrations. The lack of effect cannot be ascribed to the slowness of diffusion, because the unstirred layers were not more than 800μ thick, and the non-nervous faces had no stratum of connective tissue above them.

Depriving the non-nervous face of sodium brought about some change in the resting potential, which resulted in the development of a net potential difference across the electroplate, since the nervous face, now on the inside, was unaffected. In one of the dextrose experiments the final value of the resting potential across the non-nervous face was 62 mV (Table 6), while across the nervous face of the same electroplate it was 82 mV. A neighbouring electroplate had resting potentials of 55 and 80 mV respectively. The extracellular spaces beneath the topmost electroplates were therefore at a mean potential of +22.5 mV relative to the surface. The other dextrose experiment on the non-nervous face gave a similar but smaller result (+7 mV). This net potential difference was undoubtedly genuine, and could not be eliminated even by large movements of the electrodes. In the course of the dextrose experiments on the nervous faces, the electroplates were

never transfixed, so that no direct evidence was obtained either for or against the existence of net potentials across them. However, the figures in Table 6 show that the resting potentials did have some tendency to fall in dextrose solutions. The most obvious reason for a change in the resting potential under these conditions is that there must have been an appreciable outward passage of Cl^- ions, and perhaps of Na^+ as well. But the question was not investigated further, and no detailed explanation can be offered.

TABLE 6. The effect of sodium-free solutions on the membrane potentials

Face exposed	Solution applied	Resting potential (mV)	Action potential (mV)	Depth unstirred (μ)
Nervous	Normal Ringer	79	140	550
	Dextrose for 16 min	80	0	
	Normal for 10 min	87	135	
Nervous (3)	Normal Ringer	100 ± 3	163 ± 1	350, 500, 250
	Dextrose for 15 min	83 ± 8	0	
	Normal for 17 min	93 ± 6	127 ± 7	
Nervous (4)	Normal Ringer	83 ± 3	143 ± 2	400, 400, 750, 750
	Dextrose for 15 min	80 ± 3	c. 17 ± 7	
	Normal for 11 min	82 ± 2	130 ± 5	
Nervous	Normal Ringer	83	144	350
	Dextrose for 8 min	82	0	
	Normal for 9 min	83	138	
Nervous	Normal Ringer	95	171	500
	Dextrose for 26 min	81	90	
	Normal for 10 min	99	139	
Nervous (4)	Normal Ringer	94 ± 2	168 ± 5	—
	Choline for 15 min	65 ± 2	0	
	Normal for 15 min	65 ± 2	0	
	Normal for another 35 min	60 ± 2	0	
Nervous	Normal Ringer	83	150	550
	Choline for 13 min	73	0	
	Normal for 12 min	77	0	
Non-nervous	Normal Ringer	78	103	750
	Dextrose for 29 min	74	118	
	Dextrose for another 47 min	76	82	
Non-nervous	Normal Ringer	85	140	800
	Dextrose for 33 min	70	139	
	Dextrose for another 10 min	62	95	
Non-nervous	Normal Ringer	81	143	700
	Choline for 16 min	75	127	
	Choline for another 26 min	70	82	

For experiments in which the membrane potentials were determined in several neighbouring electroplates, the number of electroplates in the group is stated in brackets, and the standard error of the mean is given. The other experiments were performed with single electroplates. The last column shows the depth of the unstirred layer of jelly above the actual face of each electroplate. In the third experiment the two deeper electroplates did not block completely, but it was difficult to measure the residual response accurately.

In the choline experiment on the non-nervous face, a net potential of -10 mV was developed across the electroplate. Possibly enough choline had penetrated into and beyond the top electroplates to depolarize their nervous faces.

In collaboration with Prof. Chagas, a slice of main organ was on one occasion perfused with the isotonic dextrose solution, using the technique described by Chagas *et al.* (1951). After more than 2 hr of perfusion there was still no change in the size of the direct response, although the indirect response disappeared (as did that of a control slice perfused with Ringer). Calculations showed

that this negative result was attributable to the slow rate of perfusion, and that it would have taken many hours to wash out an appreciable fraction of the sodium in the very large extracellular space of the tissue. It is, of course, much easier to introduce small amounts of drugs by perfusion, than to remove substances already in the slice.

The effect of applying potassium-rich solutions

The depolarizing effect of potassium is a familiar phenomenon, which is generally accepted as evidence that the resting potential in excitable tissues arises wholly or partly from the difference in concentration of potassium ions on the two sides of the membrane (Hodgkin, 1951). It was obviously important to see whether the resting potentials of the electroplates were reduced by the application of potassium-rich solutions, but it was clear even before trying the experiment that the nature of the preparation might make it hard to obtain satisfactory results. As the potassium concentration could only be changed on one side of the electroplate at a time, it was plainly impossible to avoid setting up a net potential difference across the electroplate, which would cause a steady flow of current through it. Whichever face was depolarized, this would offset the depolarization, and would simultaneously depolarize the opposite face—thus tending to equalize the two resting potentials. Moreover, the experiments could not be done very quickly, owing to diffusion effects, and a long-lasting passage of current in either direction might be expected to have deleterious consequences. It turned out that these misgivings were not entirely unjustified.

TABLE 7. The effect of altering the external potassium concentration

Resting potential (mV)			Action potential (mV)			Duration (msec)			Depth unstirred (μ)	Time (min) for		
A	B	C	A	B	C	A	B	C		B	C	
10 mm potassium to nervous face												
{	83	73	80	156	131	146	2.8	2.2	1.9	150	15	15
	81	69	81	160	134	122	2.9	2.3	2.2	1000	15	15
	82	73	87	150	107	128	2.3	1.1	1.3	150	15	15
	73	61	72	128	113	127	1.8	1.7	1.7	650	6-18	18
25 mm potassium to nervous face												
	92	60	80	150	137	143	1.7	1.6	1.8	1100	12	15
	79	55	80	157	108	135	2.7	1.6	2.3	1000	16-24	21
10 mm potassium to non-nervous face												
{	93	71	78	132	102	124	2.3	2.0	2.3	750	15	15
	94	94	92	149	139	135	2.6	2.2	2.5	1450	15	15
	93	68	77	146	123	80	2.7	1.8	1.9	800	15	15
	78	65	78	134	110	119	1.6	1.2	1.5	600	3-19	30
	89	66	80	156	135	119	1.9	1.8	1.7	400	9	16

A, in normal Ringer at start; B, in high potassium; C, normal Ringer restored. The figures bracketed together are for electroplates in the same slice.

Table 7 shows the results of washing electroplates with solutions containing twice (10 mm) and five times (25 mm) the normal amount of potassium (the extra potassium being additional to the usual constituents of the Ringer, not

substituted for sodium). There was almost invariably a drop in the resting potential which was, as a rule, at least partially reversible. There was also a fall in the action potential, usually somewhat greater in extent, together with a reduction in its duration. The original size was seldom recovered on restoring normal Ringer, and nearly half the electroplates tested were in the end either inexcitable or giving very small spikes. The figures for eight electroplates whose condition deteriorated seriously have been omitted from the table; their resting potentials fell by rather more than the averages given below. One electroplate failed to show any effect other than a slight shortening

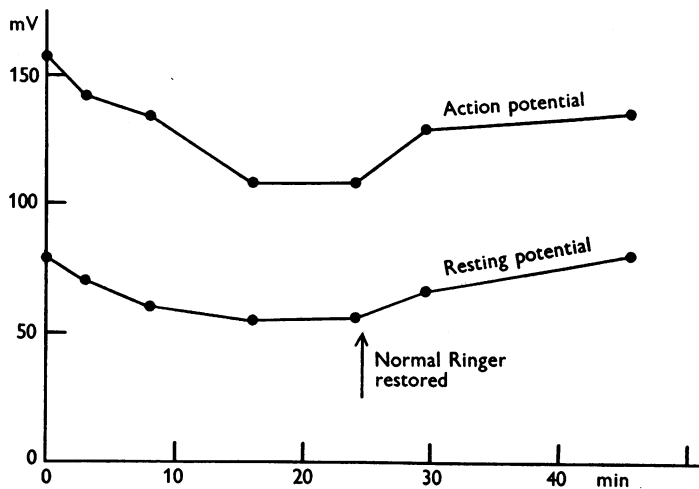


Fig. 14. The effect on the resting and action potential of washing the nervous face of an electroplate with a Ringer's solution containing five times the normal amount of potassium. Depth of unstirred layer 100μ . Temperature 25°C .

of the spike, probably because it was protected by an exceptionally deep (1450μ) unstirred layer. Fig. 14 shows the time-course of the potential changes in one of the better experiments—better in the sense of being properly reversible—and provides some indication that the times allowed both for the development of the effect, and for recovery, were adequate. This was further confirmed in three other experiments of the same type, and by calculations based on the curves given in Fig. 13.

When the potassium concentration was twice normal, the average reduction in resting potential was 10.9 mV for the nervous face and 12.4 mV for the non-nervous face. With five times the normal potassium the nervous face was depolarized by 25.3 mV . The observed drops in potential were thus appreciably smaller than the theoretical changes given by a potassium electrode ($58 \log_{10} 2$, or about 17 mV , for a doubled potassium concentration), but at these relatively low potassium concentrations other excitable tissues have been shown to

behave in the same way (see Hodgkin, 1951). The depolarizations may also have been partly suppressed by current flow. No measurements were made of the net voltages across electroplates treated with high potassium on their nervous faces, but those exposed on the other face had an average of only +2 mV across them in two of the experiments, and no measurable potentials in the remainder. This implies that there may have been appreciable current flow, or alternatively that enough potassium may have escaped into the extracellular spaces beneath the electroplates—either by diffusion or carried by current—to depolarize their inner as well as their outer faces.

As the action potentials did not recover fully in normal Ringer it was difficult to decide precisely how much they were reduced in the potassium-rich solutions. Furthermore, part of the reduction might have resulted from a rise in threshold coupled with a tendency not to give an absolutely all-or-none response (see p. 332), since the stimulus strength was not always increased during the experiment. However, in the four cases where the final recovery was most complete, the mean decrease in spike size was nearly equal to the mean decrease in resting potential, which suggests that the overshoot of the action potential was not greatly affected by changing the potassium concentration.

Propagation of the action potential along the electroplates

From the findings discussed on the preceding pages, it was clear that the mechanism which generated the action potentials of the electroplates had much in common with that responsible for the propagated activity of nerve and muscle fibres. It thus seemed possible that, given suitable conditions, the nervous face of an electroplate might be capable of conducting an impulse. This expectation was verified in several experiments of the kind illustrated in Fig. 15.

Slices were cut from the organ of Sachs in two rather large eels, at the level where the electroplates were longest. They were divided in two along the central septum, so as to leave intact all the electroplates on one side. They were then mounted in the dish with the upper silver grid covering the whole surface as usual, but with the lower silver plate displaced sideways so that it only overlapped the electroplates for 1 or 2 mm at their extreme ends. The membrane potentials were recorded in the normal way with two microelectrodes close together, at various points along the electroplates, distances being read off on the scale provided on the mechanical stage (see Fig. 1). As Fig. 15 shows, when a liminal shock was applied to the lower plate, the action potential arose with a delay which increased linearly with distance from the stimulated end.

The simplest interpretation of this experiment is that the spike was being propagated along the electroplate in a fashion directly analogous to conduction

in a nerve fibre, activity at one point affecting the next by local circuit action, and so on. But before it can be accepted, two alternative explanations must be considered. First, the stimulating current must have spread beyond that end of the electroplate at which it was applied, and it might be argued that in

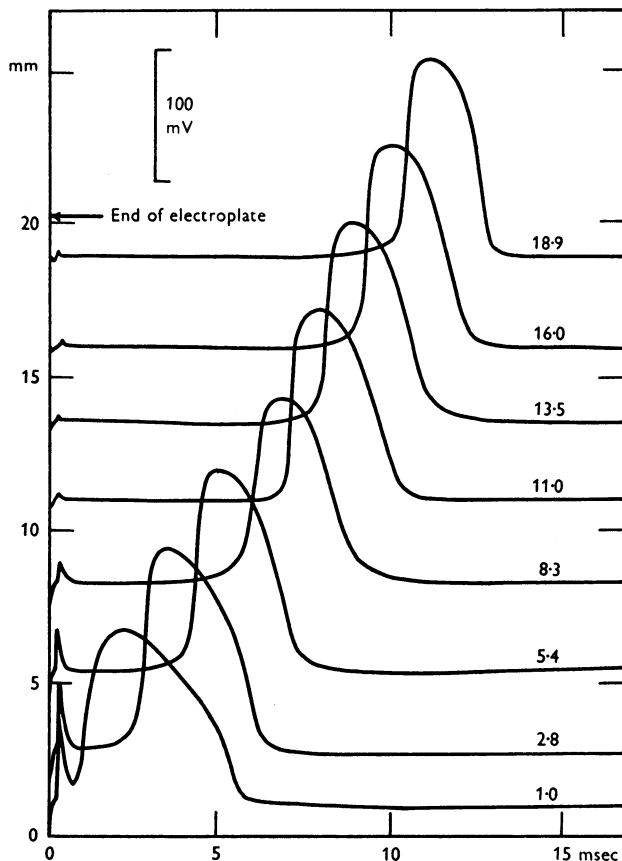


Fig. 15. Tracings showing the result of stimulating an electroplate with a liminal shock at one end only (in this case the central end), and recording through the nervous face at various points along it. Each record has been traced so that its base-line would cross the ordinate axis at a point corresponding to the distance of the electrodes from the stimulated end. The actual distances are given (in mm) against the curves. Abscissa: time in msec from beginning of stimulus artifact. The recorded resting potential varied between 80 and 86 mV at the different places. Temperature 22° C.

fact the membrane was excited in every place by the original shock, the increased delay resulting merely from progressive attenuation of the current reaching the more distant parts. However, great care was taken to set the shock strength as close as possible to its threshold value, and it is unlikely that a stimulus only just able to produce a response in the membrane im-

mediately above the lower plate could also have excited the peripheral end of the electroplate, where it must have been well below threshold. Moreover, the delay in the appearance of the spike at the far end was as much as 14.3 msec in one experiment, while the greatest delay ever observed when the whole nervous face was stimulated with a short shock was about 3 msec at threshold. Nevertheless, the spread of stimulating current could certainly have spoiled the experiment, because it was found that with a large enough shock the delay could be reduced to zero, or to any intermediate value, all the way along the electroplate.

The second possibility is that the spike might have been propagated indirectly, by excitation of the nerve endings at each point in succession. This could best have been examined, though unfortunately the test was not made, by seeing whether a curarized slice was capable of conducting an impulse. However, the indirect response was rather difficult to set up at the very low repetition rates—under 1/sec—employed in this experiment, and Altamirano-Orrego, Coates, Grundfest, Nachmansohn & Wilson (1952), working with nerve-organ preparations, have even reported single shock nerve stimulation to be relatively ineffective, though two shocks close together produced a discharge. It therefore seems improbable—but, until further work has been done, not impossible—that the spike could have been propagated as a wave of indirect responses.

The effective conduction velocity in the experiment of Fig. 15 was 2.2 m/sec. In three other experiments it was 1.4, 1.4, and 1.6 m/sec, giving an overall average of 1.7 m/sec. The lengthening of the spike at the stimulated end was observed in every case, but elsewhere in the electroplate there was not much variation in its shape or size. Except in one experiment (see legend for Fig. 15), we neglected to note which end of the electroplate was stimulated.

DISCUSSION

There has been much dispute as to the origin and localization of the electromotive force in the electric organ. Forty years ago, Bernstein (1912) put forward the idea that there might be a semi-permanent polarization of both faces of the resting electroplate, the discharge being produced by a momentary collapse of the potential across the innervated face. But the repeated failure of all efforts, using external electrodes, to detect any appreciable resting potentials, made it difficult to accept this hypothesis, and a variety of more or less plausible schemes were proposed to supplant it (for references see Fessard, 1946). The results illustrated in Figs. 4–6 now prove beyond reasonable doubt that Bernstein's suggestion was basically correct, at least for *Electrophorus*—since it remains to be seen whether or not the electric organs of *Torpedo* and other electric fishes function in the same way. It only has to be modified in that the potential across the nervous face does not just disappear, but is

actually reversed by well over 60 mV. On the question of localizing the membrane potentials, little more need be said, and the diagram in Fig. 7 can be left to speak for itself.

Because of the embryonic derivation of the electric organ from muscle, except in the case of *Malapterurus* (Biedermann, 1898), and of the extensive evidence that a cholinergic mechanism is involved in its working (Feldberg, Fessard & Nachmansohn, 1940; Feldberg & Fessard, 1942), it has become customary to regard the electroplate as a modified motor end-plate (see, for example, Prosser, 1950, and Davson, 1951). However, a comparison of the properties of the electroplates in *Electrophorus* with those of the motor end-plates in frog muscle, recently studied in detail by Fatt & Katz (1951*a*), provokes a query as to the validity of this view. In order to be strictly just, the electroplates should be compared with the end-plates in the skeletal muscle of *Electrophorus* itself, but these have never been suitably examined, and it will therefore have to be assumed that amphibian end-plates can fairly be taken as representative of other vertebrates. One of the crucial points in the conclusions reached by Fatt & Katz is that the electric response of the end-plate membrane involves a depolarization—i.e. a decrease of the membrane potential towards zero, but not an overshoot. In contrast, in the electroplates of *Electrophorus* the potential across the nervous face undergoes a reversal during activity which is probably larger than in any other excitable tissue (see Hodgkin, 1951, table 1). It follows that it may, after all, be wrong to class the electroplate as a specially adapted end-plate.

It would be unsatisfactory to raise objections to a view which has been so widely accepted, and which may yet prove to be correct in the case of *Torpedo*, if it were impossible to reconcile the facts in an alternative and straightforward way. But there seems to be no good reason for supposing that the electroplate is not simply a much altered muscle fibre. Consider, for instance, the characteristics of the nervous face: when sufficiently depolarized it responds by producing an action potential which overshoots the resting potential; it behaves in an approximately all-or-none fashion, giving what appears to be a local response for subthreshold shocks; its action potential is abolished when there are no sodium ions in the surrounding medium, while its resting potential is dependent on the external potassium concentration; its action potential is greatly lengthened by cooling, without much change in the extent of the overshoot; and, finally, it seems to be capable of conducting an impulse. All the items on this list are well-known properties of nerve and muscle fibres. Although final conclusions cannot be reached until many of them have been studied rather more quantitatively than was possible in the time at our disposal, taken together they provide moderately strong evidence that the membrane of the nervous face is essentially a muscle membrane. The behaviour of the non-nervous face is consistent with that of a muscle membrane which

has suffered the indignity of losing its excitability—a retrogressive evolutionary change which might be relatively easy to achieve. At the same time its normal resting permeability to potassium ions (and perhaps to chloride ions as well) has apparently been enhanced, so as to enable high currents to pass through it without an excessive voltage drop. It seems very likely that the development of papillae on both faces, which considerably increases their effective surface area, has the same object.

There is, of course, evidence that a release of acetylcholine plays a vital part in the discharge of the electric organ, but as far as *Electrophorus* is concerned, it can best be explained on the supposition that the nerve endings on the nervous face are cholinergic. Nachmansohn, Coates & Rothenberg (1946) have shown that in different parts of the electric organ, the voltage produced is directly proportional to the amount of cholinesterase present (which is very high) despite wide variations in the spacing of the electroplates. Since the voltage is also proportional to the number of electroplates per cm, this can be interpreted as meaning that the cholinesterase is largely intracellular or closely associated with the surfaces, and that each electroplate contains roughly the same quantity. However, it seems that cholinesterase may be involved only in the indirect response, for Chagas *et al.* (1951) have found that slices of main organ can be perfused with eserine until their esterase activity is almost zero, without any diminution of the direct response. There are other and better reasons for believing that the main role of acetylcholine in the electric organ is that of a junctional transmitter. Thus Albe-Fessard & Chagas (1951) have shown that when the main organ is perfused with curare, its indirect response is abolished, but the direct response is not affected. This observation has been confirmed by Altamirano-Orrego *et al.* (1952), and holds good for perfusion of Sachs's organ as well (Chagas, Keynes & Martins-Ferreira, unpublished). Chagas (1952) has also reported that the acetylcholine content of the main organ decreases markedly after indirect stimulation, and is resynthesized on resting, but is very little reduced by direct stimulation. It is true that, as has already been mentioned, our slices of Sachs's organ treated with acetylcholine became depolarized and inexcitable, without apparently giving an initial discharge; but this may well be an example of the paralysing effect of too much acetylcholine, and a sufficiently rapid injection of much smaller quantities really close to the nervous face might give different results.

If we are right in postulating that the response of the nervous face is fundamentally similar to the conducted spike in a muscle fibre, then the immediate source of the electrical energy dissipated in the discharge is likely to be an exchange of sodium and potassium ions across the two faces of the electroplate. Experiments on cephalopod axons and other tissues (Keynes & Lewis, 1951; Hodgkin, 1951) have shown that during activity the energy used in generating the action currents is derived from a net gain of sodium, roughly

balanced by a loss of potassium. Arguing by analogy, the principal ionic shifts in the electroplate would be an entry of sodium through the nervous face and a leakage of potassium through the non-nervous face, though some potassium would move out through the nervous face during the falling phase of the spike. This sodium-potassium exchange would be reversed during recovery by some sort of ionic pump drawing on metabolic sources for its supply of energy. It must be admitted that the lack of any direct confirmation that such ionic movements do occur in the electroplate, is a serious loophole in our thesis. It is also one which will not easily be filled, since the very large extracellular space of the electric organ will make it exceptionally hard either to obtain reliable figures for the intracellular ionic concentrations, or to measure the ionic fluxes during stimulation.

The energetics of the electric organ have been studied in a large series of eels by Nachmansohn, Coates, Rothenberg & Brown (1946). Of the total energy available from the observed breakdown of phosphocreatine and formation of lactic acid, they found that about 40 % appeared as electrical energy. This evidence conforms with that for other excitable tissues (Ling & Gerard, 1949; Keynes, 1951) in showing that energy-rich phosphate bonds are probably the primary source of the chemical energy needed for the process of restoring the resting condition after a bout of activity. It must be pointed out, however, that in the absence of proof that the resting ionic concentrations had been fully restored, the figure of 40 % only sets an upper limit to the overall efficiency of the electric organ as a generator of electricity.

It has already been remarked in the course of this discussion that the electric organs in *Torpedo* and other fishes cannot be assumed to work exactly like that in *Electrophorus*. Nevertheless, there is a natural attraction in the idea that all electric organs derived from muscle—once again *Malapterurus* has to be excepted—are similar in their mode of operation, and it is therefore interesting to see how far the information available about *Torpedo* fits in with it. At first sight, the electric organ of *Torpedo* would appear to depend wholly on a cholinergic system. Thus on curarization, or on degeneration of the afferent nerves, it becomes totally inexcitable (Fessard, 1946). It shows no directional preferences in its excitability, and will under no circumstances give an undelayed direct response as *Electrophorus* does, but only a spike which arises with a minimum latency of about 2 msec (Fessard, 1947; Albe-Fessard, 1950*a*). It contains a large amount of acetylcholine and of cholinesterase, and using eserinated perfusion fluids it has been possible to demonstrate both an electrogenic action of injected acetylcholine, and a release of acetylcholine during activity (Feldberg *et al.* 1940; Feldberg & Fessard, 1942).

On the other hand, there is some evidence that is possibly more consistent with a mechanism of the kind found in *Electrophorus*, though none of it is unequivocal. It has been shown, for example, that the resistance of the

electric organ in *Torpedo* falls during the discharge (Albe-Fessard, 1950*b*), and that although it cannot be excited directly, passage of current through it in certain directions does reduce the latency for simultaneous nerve stimulation (Albe-Fessard & Buser, 1950) and has facilitating effects (Albe-Fessard *et al.* 1949). Then Fessard & Tauc (1951) have recently found with microelectrodes that the electroplates of *Torpedo* have a resting potential averaging about 50 mV. They did not record any action potentials (the extreme thinness and close packing of the electroplates make the electric organ of *Torpedo* technically much harder to handle than *Electrophorus*), and until this has been done, only limited deductions can be made from their observation. But the spike produced by each electroplate should be about 100 mV (Albe-Fessard, 1950*a*), and as this is twice the resting potential, it is difficult to envisage a discharge mechanism which does not involve a reversal of membrane potential across the nervous face. Finally, there is the important experiment performed by Bernstein & Tschermak (1906), in which they showed that, during the discharge, the electric organ of *Torpedo* absorbs a quantity of heat roughly equal to that produced by the electric currents flowing in the internal and external resistances of the system. They regarded this, quite justifiably, as confirmatory evidence that the immediate source of energy for the discharge was an ionic exchange rather than a chemical reaction, since in the former case the movement of ions from concentrated to dilute solutions would have caused enough cooling just to offset the total heat arising from current flow, whereas in any likely chemical reaction the ratio of heat evolved to free energy change would probably not have been much less than unity (Hodgkin (1951), points out that for the conversion of ATP to ADP it is 1.2). These measurements of heat production, which should certainly be repeated for *Electrophorus*, do not, all the same, rule out the possibility that in *Torpedo* the electroplates might work by a non-selective ionic exchange through the nervous face, as suggested by Fatt & Katz (1951*a*) for motor end-plates, rather than by a specific sodium-potassium exchange.

Although the question of the discharge mechanism in *Torpedo* must, therefore, be left for future research to decide, it may be worth making a somewhat tentative suggestion to explain the differences between the two species. Biedermann (1898) states that in *Torpedo* the terminal nervous arborization closely covers the entire surface of the electroplate, while Couceiro & Akerman (1948) have shown that in *Electrophorus* the nerve endings, though widely distributed, occupy only a small fraction of the total area of the nervous face. The acetylcholine contents of the two electric organs are similarly contrasted, that of *Torpedo* being 40–100 $\mu\text{g/g}$ fresh tissue (Feldberg & Fessard, 1942), whereas *Electrophorus* contains less than 5 $\mu\text{g/g}$ (Chagas, 1952). It may be that they discharge in more or less the same manner, but that in *Torpedo* an end-plate type of membrane largely predominates over a relatively small

area of muscle-type membrane, and that in *Electrophorus* the situation is reversed.

In conclusion, of the many unsolved questions connected with the working of the electric organ, one in particular inevitably occurs to anyone pre-occupied, as we were, with the organ of Sachs. This is the problem of its physiological function. It is so sharply differentiated from the main organ by the increased spacing of its electroplates that it must surely serve some purpose other than merely reinforcing the discharge of the main organ. Coates, Cox & Granath (1937) found that the small pulse which generally precedes a group of four full-sized ones in the eel's natural discharge (Albe-Fessard, 1950*a*) seemed to originate in the more posterior part of the fish, and suggested that it might be generated exclusively in the organ of Sachs. But some doubt is cast on this idea by the observation of Albe-Fessard, Chagas, Couceiro & Fessard (1951) that the small wave persisted in the anterior segments of the main organ even after Sachs's organ had been put out of action by transecting the spinal cord above it. Coates (1950) has since ascribed an altogether different role to the organ of Sachs, proposing that it is the transmitter for a direction-finding system which enables the eel (whose eyes have degenerated) to find its way about and locate other fish. This possibility has now acquired fresh interest in view of Lissmann's (1951) discovery that *Gymnarchus niloticus* and certain other fishes (including *Gymnotus carapo*, a South American species closely related to *Electrophorus*) make use of electrical pulses to detect objects close to them in the water. We have repeated some of Lissmann's experiments, as described in a recent popular article (Keynes, 1952), and have verified that *Electrophorus* is capable of locating electrical conductors placed in the water beside it. We also confirmed that the direction-finding pulses arose from the posterior part of the electric organ. However, in experiments involving transection of the spinal cord we were unable to prove to our complete satisfaction that the organ of Sachs was the sole part of the electric organ employed as a transmitter. The problem of assigning a function to the organ of Sachs is an interesting one which deserves further attention.

SUMMARY

1. The membrane potentials in the electroplates of *Electrophorus electricus* were studied with the help of glass microelectrodes filled with 3M-KCl. In order to record the action potential faithfully, it was necessary to use two microelectrodes, and to measure the difference in potential between their tips when they were manoeuvred so as to be close together, but on opposite sides of one face of the electroplate or the other. Most of the experiments were done with slices of tissue cut from the organ of Sachs, where the electroplates are conveniently spaced some distance apart.

2. In the electroplates of the organ of Sachs there was an average resting potential, at 24°C , of 84 mV across each face, the inside of the cell being negative with respect to the outside. On stimulation the nervous (posterior) face produced a large spike, at the peak of which the membrane potential was reversed by 67 mV. The average duration of the spike, measured at half its peak amplitude, was 2.3 msec. The potential across the non-nervous face did not alter during activity, except for a passive voltage drop of a few millivolts resulting from the flow of current through it. In the main electric organ the membrane potentials were similar, but the spikes were slightly smaller and appreciably shorter.

3. In the resting condition there was therefore no net potential across the electroplates, but during activity the voltages across the opposite faces added together, each electroplate contributing 151 mV (on open circuit) to the total discharge. This discharge mechanism is in essence that originally proposed by Bernstein (1912), except that the voltage across the nervous face is reversed instead of just falling to zero.

4. On cooling, the spikes were greatly lengthened ($Q_{10}=2.5$), but their amplitude changed relatively little.

5. The condition for eliciting the direct (undelayed) response of the nervous face was to apply a brief depolarization. For shocks longer than 20 msec the critical depolarization was just under 12 mV. The spike arose in an approximately all-or-none fashion, and local responses were observed on subthreshold stimulation.

6. The effective electrical resistance of the nervous face was of the order of $10\ \Omega\text{cm}^2$. There was some evidence that it was reduced during the spike. The non-nervous face had a resistance estimated as $0.23\ \Omega\text{cm}^2$.

7. When the nervous face was treated with an isotonic dextrose solution, the action potential was reversibly abolished without any great change in resting potential. Choline solutions produced an irreversible inexcitability. When applied to the non-nervous face, sodium-free solutions only caused a rather slow decline in the size of the spike.

8. Potassium-rich solutions had a depolarizing effect on both faces of the electroplate, which was generally reversible.

9. When an electroplate was stimulated with a minimal shock at its extreme end, it appeared to be capable of conducting an impulse along its whole length, at an average velocity of 1.7 m/sec.

10. It is suggested that in *Electrophorus* the electroplates should be regarded not as modified motor end-plates, but as specially adapted muscle fibres, which have retained a normal excitable membrane on one side, and have lost their excitability on the other.

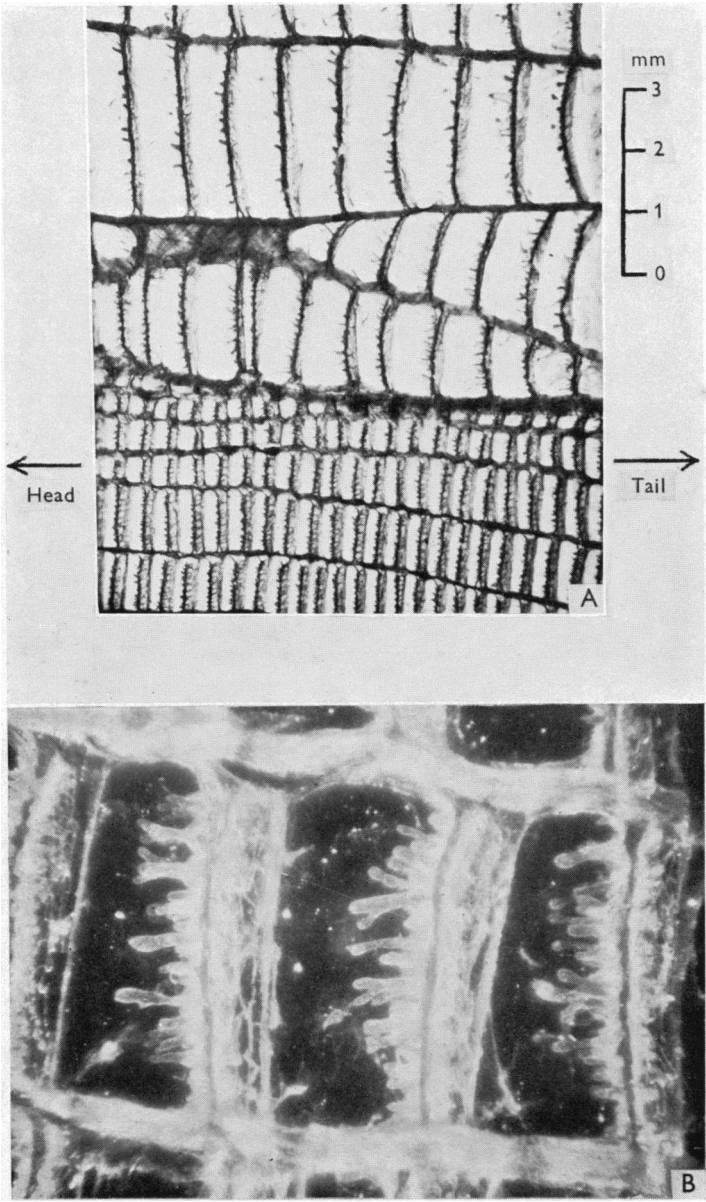
We are greatly indebted to Prof. Carlos Chagas for the many facilities placed at our disposal, and for a lavish supply of eels. R.D.K. also wishes to thank Prof. Chagas for a most welcome invitation to work in his laboratory, and the Anglo-Brazilian Society for their part in arranging the visit. Travelling expenses and cost of apparatus were met by grants from the Rockefeller Foundation, the Nuffield Foundation, and Trinity College, Cambridge. Mr A. F. Huxley kindly read and criticized the first draft of our paper.

EXPLANATION OF PLATE

Photomicrographs of sections through the electric organ of *Electrophorus*, cut in a vertical plane parallel to the spinal cord of the fish. A, stained with periodic acid and Schiff's reagent, showing the junction between the main organ and the organ of Sachs. B, a thicker section, magnification about five times greater, photographed with dark ground illumination, showing the papillae on the non-nervous face particularly well.

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